

Averox Private Ltd.

EmrSoft

Application Specifications

AI-Powered Healthcare Platform & Complete System Reference

Version 1.0 | Generated: 19 June 2026

Multi-Tenant SaaS EMR | Web + Mobile | Field-Level Encryption

This document describes the complete EmrSoft electronic medical records platform: architecture, modules, roles, database, APIs, integrations, security, and deployment configuration.

1. Executive Summary

EmrSoft is a comprehensive, multi-tenant Electronic Medical Records (EMR) and practice management platform. It serves healthcare organizations through isolated tenant subdomains, role-based portals for clinical and administrative staff, a SaaS owner console for subscription management, and patient-facing self-service features.

1.1 Platform Identity

Property	Value
Product Name	EmrSoft
Platform Title	EmrSoft- AI-Powered Healthcare Platform
Production URL	https://app.emrsoft.ai/
Demo Tenant Subdomain	emrsoft
Demo Organization	emrSoft Healthcare
Copyright	© 2025 Averox Private Ltd. All rights reserved.
Primary Logo	/EMR-Soft-Logo/emr-logo.png
Favicon / Title Logo	/EMR-Soft-Logo/emr-title-logo.png

1.2 Core Capabilities

- Patient demographics, medical records, and encrypted PHI storage
- Appointment scheduling, calendar, shifts, and telemedicine (LiveKit)
- Prescriptions, lab results, medical imaging, and clinical photography
- Billing, invoicing, insurance claims, and revenue analytics
- Inventory, pharmacy, and supply chain management
- Dynamic forms, surveys, and patient communication (email/SMS)
- AI clinical insights, chatbot, voice documentation, and decision support
- GDPR compliance, audit trails, and data subject rights
- SaaS subscription billing via Stripe with packages and trials
- QuickBooks accounting integration and PayPal payments

1.3 Technology Stack Summary

Layer	Technologies
Frontend	React 18, TypeScript, Vite, Tailwind CSS, shadcn/ui, Radix UI, Wouter, TanStack Query
Backend	Node.js, Express, TypeScript, Socket.IO, Drizzle ORM
Database	PostgreSQL (custom schema, e.g. emrsoft_encrypted)
Mobile	Flutter (Doctor App + Patient App)
Auth	JWT, bcrypt, session cookies, RBAC
Encryption	@averox/curaemrencryption-crypto-sdk (field-level patient data)
AI	OpenAI GPT, Anthropic Claude
Payments	Stripe, PayPal
Comms	Nodemailer/SMTP (Stackmail), Twilio SMS/WhatsApp

Layer	Technologies
Video	LiveKit via mk1.averox.com infrastructure

2. System Architecture

2.1 Multi-Tenant SaaS Model

A single application instance serves multiple healthcare organizations. Each organization is identified by a unique subdomain and organizationId. All data tables include tenant isolation via organizationId. The X-Tenant-Subdomain HTTP header routes requests to the correct tenant context.

- SaaS Owner (org 0): platform administration at /saas
- Tenant Organizations: clinic portals at /{subdomain}/...
- Public routes: landing, auth, legal, trial signup, form sharing
- Legacy subdomain cura is migrated to emrsoft on seed

2.2 Architecture Layers

Layer	Description
Presentation	React SPA • 80+ pages, role-based dashboards, responsive UI
API Gateway	Express server • auth, tenant middleware, 294+ REST endpoints
Business Logic	27+ domain services (AI, billing, pharmacy, GDPR, etc.)
Data Access	Drizzle ORM + storage.ts • PostgreSQL with 117+ tables
Real-time	Socket.IO, Server-Sent Events for notifications
External	Stripe, Twilio, QuickBooks, LiveKit, OpenAI, Anthropic

2.3 Request Flow

- Client sends Authorization: Bearer <JWT> and X-Tenant-Subdomain: <subdomain>
- Tenant middleware resolves organization and validates subscription status
- Auth middleware validates JWT and injects user + role context
- Route handler executes business logic with organizationId scoping
- Encrypted patient fields decrypted per-row via encryption SDK session

2.4 Project Structure

Path	Purpose
client/src/	React frontend • pages, components, hooks, services
server/	Express API • routes.ts, middleware, services, seed scripts
shared/	Drizzle schema, demo credentials, shared types
encryption-sdk/	Averox field-level encryption SDK
SQL Files/	Database seed scripts (demo users INSERT)
public/	Static assets, chatbot embed, manifest
docs/	Generated documentation PDFs

3. User Roles & Access

3.1 Supported Roles

Role	Portal Access	Primary Functions
SaaS Admin	/saas	Manage customers, packages, subscriptions, billing
Admin	Full tenant	Users, settings, billing, analytics, compliance
Doctor	Clinical	Patients, consultations, prescriptions, imaging
Nurse	Clinical	Patient care, appointments, records, messaging
Lab Technician	Lab	Lab results processing and dashboard
Sample Taker	Lab	Sample collection workflows
Pharmacist	Pharmacy	Prescription fulfillment, inventory, reports
Patient	Portal	Appointments, records, prescriptions, messaging
Receptionist	Front desk	Scheduling, registration (where enabled)

3.2 Demo Credentials (Development)

Login: /auth/login | SaaS: /saas | Header: X-Tenant-Subdomain: emrsoft

Role	Email	Password
Doctor	paul@emrsoft.ai	doctor123
Nurse	emma@emrsoft.ai	nurse123
Patient	john@emrsoft.ai	patient123
Lab Technician	amelia@emrsoft.ai	lab123
Sample Taker	sampletaker@emrsoft.ai	sample123
Pharmacist	Pharmacist@emrsoft.ai	pharmacist123
Admin	james@emrsoft.ai	467fe887
SaaS Admin	saas_admin@emrsoft.ai	admin123

3.3 Authentication

• ½ JWT tokens with 7-day expiry; login via email or username

• ½ Password hashing: bcrypt (12 salt rounds)

• ½ Password reset via email token flow

• ½ Trial signup with email verification at /create-trial

• ½ Role hierarchy enforces endpoint and UI access control

4. Application Modules & Pages

4.1 Clinical & Patient Care

Module	Route	Description
Dashboard	/:subdomain/dashboard	Role-specific KPIs and quick actions
Patients	/:subdomain/patients	Patient search, profiles, encrypted records
Calendar	/:subdomain/calendar	Appointments, scheduling, availability
Prescriptions	/:subdomain/prescriptions	E-prescribing, sharing, pharmacy handoff
Lab Results	/:subdomain/lab-results	Orders, results, reporting, PDF export
Imaging	/:subdomain/imaging	Radiology studies and DICOM-style workflows
Telemedicine	/:subdomain/telemedicine	LiveKit video consultations
Consultation	Embedded UI	Full consultation interface with body map
Clinical Procedures	/clinical-procedures	Procedure documentation
Emergency Protocols	/emergency-protocols	Protocol reference and workflows

4.2 Administrative & Financial

Module	Route	Description
Billing	/:subdomain/billing	Invoices, payments, insurance, claims
Analytics	/:subdomain/analytics	Financial and operational dashboards
Inventory	/:subdomain/inventory	Stock, POs, warehouses, alerts
Pharmacy	/:subdomain/pharmacy	Dispensing, reports, pharmacy invoices
QuickBooks	/:subdomain/quickbooks	Accounting sync and OAuth
User Management	/:subdomain/users	Staff accounts and roles
Shifts	/:subdomain/shifts	Staff scheduling and holiday settings
Settings	/:subdomain/settings	Clinic branding, headers, footers, theme
Subscription	/:subdomain/subscription	Tenant plan and payment methods

4.3 AI, Automation & Communication

Module	Route	Description
AI Insights	/:subdomain/ai-insights	Automated clinical insights
AI Agent	/:subdomain/ai-agent	Conversational AI assistant
Chatbot	/:subdomain/chatbot	Patient-facing embeddable chatbot
Voice Documentation	/voice-documentation	Speech-to-text clinical notes
Clinical Decision Support	/clinical-decision-support	Evidence-based guidance
Messaging	/:subdomain/messaging	Bulk patient SMS/email campaigns
Automation	/:subdomain/automation	Workflow automation rules

Module	Route	Description
Notifications	/:subdomain/notifications	In-app notification center
Forms	/:subdomain/forms	Form builder, sharing, responses

4.4 SaaS Portal

Module	Route	Description
SaaS Login	/saas	Platform owner authentication
SaaS Dashboard	/saas/dashboard	Customer overview and metrics
Customers	/saas/customers	Tenant organization management
Packages	/saas/packages	Subscription plan definitions
Billing	/saas/billing	Platform-level invoicing
Settings	/saas/settings	Global platform configuration

4.5 Public & Legal

🔗 Landing pages: /, /features, /pricing, /about, /help

🔗 Legal: /privacy, /terms, /gdpr, /press

🔗 Auth: /auth/login, /auth/reset-password, /patient-register

🔗 Trial: /create-trial, verify, set-password flows

🔗 Tech spec export: in-app PDF generation page

5. Database Schema

PostgreSQL database managed via Drizzle ORM. Schema name configured via DB_SCHEMA (e.g. emrsoft_encrypted). Bootstrap via server/ensure-db-schema.ts and npm run db:push.

5.1 Schema Domains (117+ tables)

Domain	Key Tables
SaaS Platform	saas_owners, saas_packages, saas_subscriptions, saas_payments, saas_invoices, saas_settings
Tenancy	organizations, clinic_headers, clinic_footers, roles, users
Patients	patients, patient_import_staging, patient_import_audit, patient_communications
Clinical	medical_records, consultations, prescriptions, lab_results, medical_images, radiology_images
Scheduling	appointments, staff_shifts, doctor_default_shifts, organization_holidays
Billing	invoices, payments, claims, insurance_payments, revenue_records, subscriptions
Inventory	inventory_items, inventory_sales, inventory_purchase_orders, inventory_warehouses, inventory_batches
Forms	forms, form_fields, form_sections, form_shares, form_responses
GDPR	gdpr_consent, gdpr_data_requests, gdpr_audit_trail, gdpr_processing_activities
AI & Analytics	ai_insights, risk_assessments, forecast_models, financial_forecasts
Integrations	quickbooks_tokens, chatbot_sessions (and related)

5.2 Data Protection

• organizationId on all tenant-scoped tables • no cross-tenant access

• Patient PHI encrypted at field level via Averox encryption SDK

• Search hashes for encrypted patient name/email lookup

• JSONB columns for flexible metadata (permissions, settings, medical history)

• Audit trails for GDPR and compliance reporting

5.3 Seeding & Demo Data

• server/ensure-saas-organization-seed.ts • idempotent org + users + subscription

• server/ensure-demo-users.ts • demo role accounts

• server/ensure-saas-admin.ts • SaaS owner account (org 0)

• SQL Files/emrsoft_demo_users_insert.sql • manual INSERT with pgcrypto

• POST /api/production-demo-setup • production bootstrap endpoint

6. API Specification

RESTful JSON API served from Express. All authenticated tenant requests require JWT and X-Tenant-Subdomain headers. 294+ endpoints organized by domain in server/routes.ts.

6.1 Required Headers

Header	Value	Required
Authorization	Bearer <jwt_token>	Authenticated routes
X-Tenant-Subdomain	emrsoft (demo)	Tenant-scoped routes
Content-Type	application/json	POST/PATCH/PUT bodies

6.2 Core Endpoint Groups

Group	Examples	Notes
Authentication	POST /api/auth/login, GET /api/auth/me	Public login; JWT response
SaaS	POST /api/saas/login, GET /api/saas/customers	Platform owner only
Patients	GET/POST/PATCH /api/patients	Encrypted field handling
Appointments	GET/POST/PATCH /api/appointments	Calendar integration
Prescriptions	GET/POST /api/prescriptions	Pharmacy workflow
Lab Results	GET/POST /api/lab-results	PDF generation, sharing
Billing	GET/POST /api/invoices, /api/payments	Stripe webhooks
Inventory	GET/POST /api/inventory/*	Stock management
Forms	GET/POST /api/forms, /api/form-shares	Public form fill URLs
Chatbot	POST /api/chatbot/message	API key + org ID auth
GDPR	GET/POST /api/gdpr/*	Data subject requests
Dashboard	GET /api/dashboard/*	Role-specific stats
Telemedicine	POST /api/livekit/*	Room creation via LiveKit

6.3 Response Conventions

🔗 ½ Success: JSON object with entity data or { message: 'Success' }

🔗 ½ Errors: { error: string, code?: string, details?: array } with HTTP 4xx/5xx

🔗 ½ Pagination: limit/offset or cursor on list endpoints where applicable

🔗 ½ File uploads: multipart/form-data via Multer middleware

7. Backend Services

Service	File	Responsibility
Authentication	services/auth.ts	JWT issue/verify, password hashing
Email	services/email.ts	SMTP templates i.e welcome, reset, reminders
AI	services/ai.ts	OpenAI integration, clinical NLP, drug interactions
Chatbot AI	services/chatbot-ai.ts	Embeddable patient chatbot logic
Anthropic	anthropic.ts	Claude clinical assistant
GDPR	services/gdpr-compliance.ts	Consent, data requests, audit trail
Appointments	services/appointment-scheduler.ts	Scheduling rules and conflicts
Prescriptions	services/prescription-management.ts	Medication workflows
Clinical CDS	services/clinical-decision-support.ts	Decision support rules
Inventory	services/inventory.ts	Stock, sales, purchase orders
Pharmacy	services/pharmacy.ts	Dispensing and pharmacy reports
Stripe	services/stripe-*.ts	Subscriptions, payments, webhooks
QuickBooks	services/quickbooks/*	OAuth, API sync, webhooks
Twilio	services/twilio-connector.ts	SMS and WhatsApp delivery
SMS Scheduler	services/sms-scheduler.ts	Scheduled message dispatch
Subscription Reminders	services/subscription-reminders.ts	Billing reminder emails
Patient Import	services/patient-import.ts	Bulk patient CSV import
Health Dashboard	services/health-dashboard.ts	Aggregated dashboard metrics
Audit	services/audit-compliance.ts	Compliance audit logging

7.1 Middleware

i.e tenant.ts i.e subdomain resolution, subscription validation

i.e auth i.e JWT validation, user context injection

i.e role checks i.e per-route RBAC enforcement

i.e Multer i.e file upload handling for images and documents

8. Security, Encryption & Compliance

8.1 Field-Level Encryption

Patient sensitive data is encrypted using @averox/curaemrencryption-crypto-sdk. Each patient record has a data encryption key (DEK) wrapped with organization secrets. Encryption constants (cura-emr:patient-record:v1) are stable identifiers & do not change in production without migration.

• server/utls/encryption-sdk.ts • encrypt/decrypt helpers

• server/utls/patient-search-hashes.ts • blind index for search

• Row-level encryption sessions per patient access

8.2 Authentication Security

• JWT_SECRET must be set in production (never use fallback default)

• bcrypt password hashing with 12 rounds

• Password reset tokens with expiry

• Session cookies for web auth alongside JWT

8.3 GDPR Compliance

• Data subject rights: access, rectification, erasure, portability

• Consent tracking with withdrawal support

• Processing activity records and audit trails

• Dedicated GDPR compliance pages and API endpoints

8.4 Production Security Checklist

Item	Environment Variable / Action
JWT secret	JWT_SECRET=<strong-random-value>
Database	DATABASE_URL, DB_SCHEMA, DB_USER, DB_PASSWORD
SMTP	SMTP_HOST, SMTP_USER, SMTP_PASSWORD, EMAIL_FROM
Stripe	STRIPE_SECRET_KEY, STRIPE_WEBHOOK_SECRET
OpenAI	OPENAI_API_KEY
Twilio	TWILIO_ACCOUNT_SID, TWILIO_AUTH_TOKEN
QuickBooks	QUICKBOOKS_CLIENT_ID, QUICKBOOKS_REDIRECT_URI
HTTPS	TLS termination at reverse proxy / load balancer
Secrets	Never commit .env; rotate credentials regularly

9. Third-Party Integrations

9.1 Payments & Billing

Provider	Use Case	Configuration
Stripe	SaaS subscriptions, tenant billing, payment methods	STRIPE_SECRET_KEY, webhook endpoints
PayPal	Alternative payment capture and refunds	PayPal SDK env vars
QuickBooks	Accounting sync, invoice export	OAuth redirect: app.emrsoft.ai

9.2 Communications

Provider	Use Case	Configuration
Stackmail SMTP	Transactional email (noreply@emrsoft.ai)	SMTP_HOST=smtp.stackmail.com:465
Twilio	SMS reminders, WhatsApp messaging	TWILIO_* env vars
SendGrid	Alternative email provider (optional)	SENDGRID_API_KEY

9.3 AI & Video

Provider	Use Case	Configuration
OpenAI	Clinical insights, NLP, documentation	OPENAI_API_KEY
Anthropic Claude	Alternative AI clinical assistant	ANTHROPIC_API_KEY
LiveKit	Telemedicine video rooms	LIVEKIT_SERVER_URL, mk1.averox.com API

9.4 Embeddable Chatbot

External sites embed EmrSoft chatbot via `public/chatbot-embed.js`. Configuration: `window.EmrSoftChatbot = { organizationId, apiKey, title, primaryColor, position, welcomeMessage }`. Container element: `<div id="emrsoft-chatbot"></div>`. Legacy CuraChatbot aliases supported for backward compatibility.

10. Frontend Architecture

10.1 Stack & Patterns

- React 18 + TypeScript compiled by Vite
- Wouter for client-side routing with subdomain-aware paths
- TanStack Query for server state, caching, and mutations
- shadcn/ui + Radix UI + Tailwind CSS design system
- React Hook Form + Zod for form validation
- Theme: emrsoft-theme with light/dark mode (CSS variables --emrsoft-*)

10.2 Branding Assets

Asset	Path
Primary Logo	/EMR-Soft-Logo/emr-logo.png
Title Logo	/EMR-Soft-Logo/emr-title-logo.png
Dark Mode Logo	/EMR-Soft-Logo/logo-Emrsoft-night.png
Favicon	/EMR-Soft-Logo/emr-title-logo.png

10.3 Key Component Areas

- components/auth/ • login, demo credentials panel, password reset
- components/dashboards/ • role-specific dashboard widgets
- components/consultation/ • full consultation interface
- components/forms/ • form builder, fill, and share UI
- components/integrations/ • QuickBooks, Stripe config panels
- pages/saas/ • SaaS owner portal components

11. Deployment & Operations

11.1 Build & Run Commands

Command	Purpose
npm install	Install deps + build encryption SDK
npm run dev	Development server (tsx server/index.ts + Vite HMR)
npm run build	Production build (Vite client + esbuild server bundle)
npm start	Run production server from dist/
npm run db:push	Sync Drizzle schema to PostgreSQL
npm run db:demo-users-sql	Generate demo user INSERT SQL
npm run docs:emrsoft-spec-pdf	Generate this specifications PDF

11.2 Environment Variables (Key)

Variable	Purpose
DATABASE_URL	PostgreSQL connection string
DB_SCHEMA	PostgreSQL schema name (e.g. emrsoft_encrypted)
BASE_URL	Public app URL (https://app.emrsoft.ai/)
JWT_SECRET	JWT signing secret
FORCE_SEED	Force re-seed demo data on startup
SMTP_* / EMAIL_FROM	Email delivery configuration
STRIPE_*	Payment processing
OPENAI_API_KEY	AI features
LIVEKIT_SERVER_URL	Video telemedicine
QUICKBOOKS_REDIRECT_URI	QuickBooks OAuth callback

11.3 Database Bootstrap

- 🔗 ½ server/ensure-db-schema.ts creates schema if missing
- 🔗 ½ Auto-runs drizzle-kit push when organizations table absent
- 🔗 ½ Seed runs on startup when FORCE_SEED=true or empty database
- 🔗 ½ Migrates legacy cura subdomain to emrsoft automatically

11.4 Mobile Applications

Flutter-based Doctor App and Patient App connect to the same REST API. Mobile clients require JWT auth and X-Tenant-Subdomain header. Features include appointments, records, prescriptions, lab results, push notifications, and offline sync.

12. Appendix

12.1 NPM Dependencies (Selected)

Package	Version	Purpose
react	^18.3.1	UI framework
express	^4.21.2	HTTP server
drizzle-orm	^0.39.1	Database ORM
stripe	^18.5.0	Payments
openai	^4.52.0	AI
@anthropic-ai/sdk	^0.37.0	Claude AI
livekit-client	^2.16.0	Video calls
socket.io	^4.8.1	Real-time events
twilio	^5.3.2	SMS/WhatsApp
jspdf	^3.0.2	PDF generation
bcrypt	^6.0.0	Password hashing
jsonwebtoken	^9.0.2	JWT auth

12.2 Document History

Version	Date	Changes
1.0	19 June 2026	Initial EmrSoft application specifications & full platform reference

© 2025 Averox Private Ltd. All rights reserved. EmrSoft & AI-Powered Healthcare Platform.

Confidential & for authorized stakeholders only.